

LUCIDPREP

Grade 11 Mathematics

U.S. Standards-Aligned Course Syllabus

ALGEBRA 2

Algebra 2 · Common Core

36 weeks • 4 modules • 70% mastery

Curriculum Architecture

This course is organized as 36 weeks across four 9-week modules. Every week follows LucidPrep's established secondary learning cycle: AI Instructor opening, objectives, vocabulary, five warm-ups, main lecture, two short tutorial videos, AI Instructor presentation, two homework sets, and a 40-question assessment.

Mastery standard: 70%+. Students below mastery review the instructional presentation and may retake the assessment, matching the existing LucidPrep secondary model.

U.S. Alignment Note

Algebra 2 is the standard Grade 11 mathematics course in the most common sequence (Algebra 1 → Geometry → Algebra 2). Integrated Mathematics states reach the same standards through Math III. This syllabus uses the Common Core high-school Algebra, Functions, Number and Statistics categories as the primary reference and cross-checks against California and Texas TEKS.

Prepared for LucidPrep • Curriculum Planning Edition • 2026

Common Instructional & Assessment Model

Weekly Learning Cycle

Step	Component	Description
1	Opening / Domain Overview	~2-minute AI Instructor introduction
2	Objectives + Vocabulary	Student-facing outcomes and core terminology
3	Warm-Up	5 questions with explanations
4	Main Lecture	Concept instruction, models and examples
5	Video Tutorials	2 short instructional videos
6	AI Instructor Presentation	Visual models, examples, strategies and applications
7	Homework	Two sets of 5 questions
8	Weekly Assessment	40 questions; 70% mastery
9	Review / Retake	Below mastery → review presentation → retake

Assessment Philosophy

Assessment is continuous and mastery-based. Each week pairs low-stakes practice (five warm-ups and two homework sets) with a 40-question weekly assessment. Students scoring below 70% review the instructional presentation and retake, so mastery is reached before progressing. Module assessments at Weeks 9, 18 and 27 and a cumulative final at Week 36 measure retention and synthesis.

ALGEBRA 2

Grade 11 mathematics aligned to the Common Core State Standards

This 36-week course is designed around **the Common Core high-school Algebra 2 content: The Complex Number System (N-CN), Polynomials & Rational Expressions (A-APR), Reasoning with Equations (A-REI), Building & Interpreting Functions (F-IF, F-BF, F-LE), Trigonometric Functions (F-TF) and Statistics & Inference (S-ID, S-IC)**. It is a full-year curriculum with four nine-week modules and a consistent LucidPrep mastery workflow.

Grade	Grade 11
Course	Algebra 2 · Common Core
Duration	36 weeks
Structure	4 modules × 9 weeks
Weekly assessment	40 questions
Mastery threshold	70%
Module assessments	Weeks 9, 18, 27
Final assessment	Week 36

Course Learning Outcomes

- Extend the number system to complex numbers and use them to solve equations.
- Analyze polynomial, rational, radical, exponential and logarithmic functions.
- Model periodic phenomena with trigonometric functions.
- Build, transform and interpret functions across representations.
- Reason with data, distributions and statistical inference.

ALGEBRA 2

Module 1 — Functions, Quadratics & Complex Numbers

Week	Topics	Students will be able to	Standards
1 — Functions Review & Transformations	Parent functions; shifts, stretches, reflections	Transform and interpret functions	F-BF.3; F-IF.7
2 — Quadratic Functions & Modeling	Vertex/standard/factored forms; key features	Analyze and model with quadratics	F-IF.8; A-SSE.3
3 — Solving Quadratics	Factoring, completing the square, quadratic formula	Solve quadratics by all methods	A-REI.4
4 — Complex Numbers	i , $i^2 = -1$, operations	Operate with complex numbers	N-CN.1/2
5 — Complex Solutions	Discriminant; complex roots	Solve quadratics with complex solutions	N-CN.7; A-REI.4b
6 — Systems Revisited	Linear-quadratic and nonlinear systems	Solve systems including nonlinear	A-REI.7
7 — Piecewise & Absolute-Value Functions	Definitions and graphs	Analyze piecewise and absolute-value functions	F-IF.7b
8 — Inverses of Functions	Finding and verifying inverses	Find and interpret inverse functions	F-BF.4
9 — Module 1 Assessment	Cumulative review	Demonstrate mastery of Module 1	N-CN/A-REI

Module delivery notes. Deepen the function families from Algebra 1 and connect them through structure, transformation and modeling. Emphasize equivalence, multiple representations and interpretation of parameters.

ALGEBRA 2

Module 2 — Polynomial, Rational & Radical Functions

Week	Topics	Students will be able to	Standards
1 — Polynomial Operations	Add, subtract, multiply; structure	Operate on polynomials	A-APR.1
2 — Factoring & Roots	Higher-degree factoring; Remainder/Factor theorems	Factor and find roots	A-APR.2/3
3 — Graphing Polynomials	End behavior, zeros, multiplicity	Graph polynomial functions	F-IF.7c
4 — Division & the Fundamental Theorem	Long/synthetic division; FTA	Divide polynomials and count roots	A-APR.2; N-CN.9
5 — Rational Expressions	Simplify, multiply/divide, add/subtract	Operate on rational expressions	A-APR.6/7
6 — Rational Functions & Equations	Asymptotes; solving; extraneous solutions	Analyze and solve rational functions	A-REI.2; F-IF.7d
7 — Radicals & Rational Exponents	Simplifying; equivalence	Rewrite radical and rational-exponent expressions	N-RN.1/2
8 — Radical Functions & Equations	Graphs; solving; extraneous roots	Analyze and solve radical equations	A-REI.2
9 — Module 2 Assessment	Cumulative review	Demonstrate mastery of Module 2	A-APR

Module delivery notes. Deepen the function families from Algebra 1 and connect them through structure, transformation and modeling. Emphasize equivalence, multiple representations and interpretation of parameters.

ALGEBRA 2

Module 3 — Exponential, Logarithmic Functions & Sequences

Week	Topics	Students will be able to	Standards
1 — Exponential Functions	Growth/decay; base e	Model with exponential functions	F-LE.1/2; F-IF.8b
2 — Logarithms	Definition; log as inverse of exponential	Evaluate and interpret logarithms	F-BF.5
3 — Properties of Logarithms	Product/quotient/power rules; change of base	Apply logarithm properties	F-LE.4
4 — Solving Exponential & Log Equations	Using logs; applications	Solve exponential and log equations	F-LE.4
5 — Modeling with Exp/Log	Compound interest, half-life, real data	Model real phenomena	F-LE.5
6 — Arithmetic & Geometric Sequences	Explicit and recursive forms	Represent and analyze sequences	F-BF.2
7 — Series & Summation	Arithmetic/geometric series; sums	Compute series sums	A-SSE.4
8 — Comparing Function Families	Linear vs exponential vs polynomial growth	Compare growth of function types	F-LE.3
9 — Module 3 Assessment	Cumulative review	Demonstrate mastery of Module 3	F-LE

Module delivery notes. Deepen the function families from Algebra 1 and connect them through structure, transformation and modeling. Emphasize equivalence, multiple representations and interpretation of parameters.

Module 4 — Trigonometry & Statistics

Week	Topics	Students will be able to	Standards
1 — Angles & the Unit Circle	Radian measure; the unit circle	Use radian measure and the unit circle	F-TF.1/2
2 — Trigonometric Functions	Sine, cosine, tangent as functions	Evaluate trigonometric functions	F-TF.2
3 — Graphing Trig Functions	Amplitude, period, phase shift	Graph and transform trig functions	F-TF.5; F-IF.7e
4 — Trigonometric Identities	Pythagorean identity; modeling periodicity	Use identities and model periodic phenomena	F-TF.8/5
5 — Data Distributions	Shape, center, spread; the normal distribution	Describe distributions; use the normal model	S-ID.4
6 — Sampling & Study Design	Surveys, experiments, observational studies	Distinguish study types and evaluate design	S-IC.3
7 — Making Inferences	Sampling variability; margin of error; simulation	Draw inferences from samples	S-IC.4
8 — Evaluating Reports & Decisions	Interpreting and evaluating statistical claims	Evaluate statistical reports and decisions	S-IC.5/6
9 — Final Assessment	Cumulative final across all modules	Demonstrate mastery of the full course	N-CN-S-IC

Module delivery notes. Deepen the function families from Algebra 1 and connect them through structure, transformation and modeling. Emphasize equivalence, multiple representations and interpretation of parameters.

Assessment & Alignment

Assessment Blueprint

Warm-ups: weekly • Homework: weekly • Weekly assessment: 40 questions • Module assessments: Weeks 9 / 18 / 27 • Final: Week 36 • Below 70%: instructional review and retake.

Standards Alignment

Primary framework	CCSS: N-CN, A-APR, A-REI, F-IF, F-BF, F-LE, F-TF, S-ID, S-IC
Mathematical Practices	Make sense of problems and persevere; reason abstractly and quantitatively; construct viable arguments and critique reasoning; model with mathematics; use tools strategically; attend to precision; use structure and regularity.
Recurring Concepts	Multiple representations; equivalence; proportional reasoning; structure; modeling; real-world application.
State mapping	Common Core adopted by the large majority of states; Texas TEKS and other state frameworks cover the same grade-level content under different codes.

Content Production Guidance

Author each week as 5 warm-up items, a concept lecture with worked examples and multiple representations, 2 curated videos, an AI presentation with models and strategies, 2 homework sets (5 items each), and a 40-question bank spanning fluency, application and reasoning. Tag every item to its Common Core code.

Curriculum Basis & Source Notes

This Grade 11 mathematics syllabus was generated from the U.S. Common Core State Standards for Mathematics. Content and pacing reflect Algebra 2, the typical Grade 11 course in the most common U.S. sequence.

Primary reference	Common Core State Standards for Mathematics — high-school Algebra, Functions, Number & Quantity and Statistics categories (N-CN, A-APR, A-REI, F-IF, F-BF, F-LE, F-TF, S-ID, S-IC) and the Standards for Mathematical Practice.
State cross-check	California Common Core and integrated-pathway (Math III) standards.
State cross-check	Texas Essential Knowledge and Skills (TEKS) for Algebra II.
LucidPrep alignment	Mapped to LucidPrep's 36-week, 4-module structure with 70% mastery.

Recommended next production step. Commission item banks per week (40 assessment items plus warm-ups and homework), each tagged to its Common Core code, then load into the LucidPrep authoring pipeline.